

This is a guide to how to think about the midterm. It is not a literal practice exam but mastering these concepts will be a good preparation for the exam. The midterm exam will be **closed book** with **no calculators**. The exam will be **75 minutes**. The midterm will cover up to and including Section 3.2 of the textbook.

1. PRACTICE QUESTIONS

Question 1. *Problems involving solutions to linear congruences and/or systems of linear congruences. Find the least possible solutions to these congruences, or state why there is no solution. Show all work.*

- $84x \equiv 5 \pmod{49}$
- $12x \equiv 16 \pmod{32}$
- $7x \equiv 1 \pmod{24}$ (*i.e. find the multiplicative inverse of 7 modulo 24.*)
- *System:*

$$x \equiv 3 \pmod{7}$$

$$x \equiv 4 \pmod{6}$$

- *Integers a, b such that $(a, b) = 3$ and $a + b = 65$.*
- *Find a reduced residue system modulo 18.*
- *Is $\{-15, 66, -75, 11\}$ a reduced residue system modulo 8? Explain.*
- *Find a reduced residue system modulo 7 that only consists of odd integers.*

Question 2. *Some Theorems to know by name and how they can be applied to solve problems.*

- *Euclidean Algorithm*
- *Chinese Remainder Theorem*
- *Fermat Little Theorem **
- *Wilson's Theorem **
- *Euler's Theorem **
- *Euclid's Theorem on Infinitude of Primes */Dirichlet's theorem for special cases (like the one proved in class) **

*An asterisk indicates that a possible question is "State and Prove Theorem *."*

Question 3. *Definition and example style question:*

- *State the definition of a multiplicative arithmetic function. State the definition of a completely multiplicative arithmetic function.*

- Prove that the Euler ϕ -function is multiplicative, but not completely multiplicative.

Question 4. These questions require the use of one of the named theorems above. Correct solutions should cite the theorem used and explain the computation/steps.

- Use the Euclidean Algorithm to find $(831, 402)$, and express it as an integer linear combination of 831 and 402.
- Prove that for odd primes p , we have $2(p-3)! \equiv -1 \pmod{p}$
- Let $a, m \in \mathbb{Z}$ and $m > 1$. If $(a, m) = 1$, show that $a^{\phi(m)-1}$ is the multiplicative inverse of a modulo m .
- Find the least positive solution x to the congruence $x \equiv 20^{110} \pmod{17}$. For $x \equiv 38^{110} \pmod{21}$.

Question 5. Some theory type questions, drawn from the key steps of proofs of theorems, propositions and/or lemmas. There are several examples below. (* denotes that it is drawn directly from our text, or a key part of the proof of one of our Theorems.)

A good exercise is to pinpoint these yourself from exercises, homework, and results in the book. I encourage to search for other examples yourself.

- * Let m be an integer greater than one. And let $\{r_1, \dots, r_{\phi(m)}\}$ be a reduced residue system modulo m . Characterize those integers j for which $\{jr_1, \dots, jr_{\phi(m)}\}$ is also a reduced residue system modulo m .
- * If $a, b \in \mathbb{Z}$, $a \geq b > 0$, and $a = bq + r$ for integer r , show that $(a, b) = (b, r)$.
- If $a, b \in \mathbb{Z}$, and $(a, b) = [a, b]$, what can you say about integers a, b ?
- * Let $a, b \in \mathbb{Z}$ and p be prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.
- * Let p be a prime, and a an integer with $p \nmid a$. Show that there is a unique solution to $ax \equiv 1 \pmod{p}$.
- Let $a, b, c, m \in \mathbb{Z}$ with $m, c \geq 1$. If $a \equiv b \pmod{m}$ then $ac \equiv bc \pmod{mc}$.
- Let $a, b, c, d \in \mathbb{Z}$, with $b, d \neq 0$, $(a, b) = (c, d) = 1$. If $\frac{a}{b} + \frac{c}{d}$ is an integer, conclude that $b = \pm d$.